Executive Summary

Title:
TikTok Project: Predicting Verified vs. Unverified Accounts

Subtitle:
Logistic Regression Modeling and Evaluation Metrics Results

Project Overview:
In this phase of the project, an exploratory data analysis (EDA) and regression modeling were applied to analyze TikTok video characteristics and their relationship with account_status. This analysis aims to reveal how specific video features relate to verified users, which will inform the final classification model for predicting whether a video presents a claim or opinion.

Key Insights:
-Model Performance: The logistic regression model achieved an accuracy of 65%. Although the precision for predicting unverified accounts was slightly lower at 61%, the model's recall of 85% for unverified accounts was higher, indicating that the model was able to catch 85% of the unverified accounts that were actually unverified.

- Video Duration: The model's coefficient uncovered that video duration (video_duration_sec) has the strongest positive association with verified status. That means for every additional second in video duration, the log odds of the account being verified increase by 0.009.

- Low impact of Engagement Metrics: Other features, such as view counts, share counts, download count, and comment counts, yield small coefficients. This indicates that they have a minimal association with the account verification status in this model.

Details:
-Methodology: Exploratory Data Analysis (EDA), data_upsampling(to address class imbalance), feature encoding (One-Hot-Encoding), logistic regression model_building using python (scikit_learn).

-Evaluation_metrics: Precision, Recall, F1-Score, Accuracy, and confusion-matrix analysis.

Variables Test:
Dependent variable: verified_status (categorical)

Independent variables: video_duration_sec, video_view_count, video_share_count, video_download_count, video_dowmload_count, video_comment_count, claim_status, and author_ban_status.

Next Steps:
- Model Integration: Incorporate video_duration_sec as a key feature in the upcoming machine learning classification model to predict whether a video contains a claim or an opinion, given its correlation with a verified account.

- Evaluate Algorithmic Bias: Conduct regular audits on models relying heavily on video duration to ensure the algorithm does not unintentionally marginalize creators who produce short-form content.

- Further Model Refinement: Since the overall accuracy of 65% is lower than what is generally considered acceptable, explore more complex classification models, such as Random Forest or XGBoost, to improve predictive performance.